LESSON PLAN

FUEL, OIL, LUBRICANT AND GREASES (28)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	13	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Extrusion, exudation and radio hazard
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Extrusion, exudation and radio hazard
5. Specific Objectives	28.1.1.1 Trainee will be able to explain the need for explosive regulation

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of explosive storage regulations and identification of extrusion/exudation signs in lubricant containers; Instructor supervision required for safety checks.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of explosive regulation?
(b) Define extrusion in the context of lubricants.
(c) Why is radio hazard a concern for aircrew?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Extrusion • Causes of Lubricant Extrusion • Definition of Exudation • Differences between Extrusion and Exudation • Impact on Aircraft Systems	L & Ds	PPT, IB	15 min
(b)	• Radioactive Contamination Sources • Types of Radioactive Hazards • Detection Methods for Radioactivity • Safety Protocols for Radioactive Materials • Regulatory Framework for Explosives	L & Ds	PPT, PDF	20 min
(c)	• Consequences of Ignoring Regulations • Maintenance Procedures for Contaminated Parts • Reporting Procedures for Hazards • Review of Explosive Storage Rules • Summary of Key Safety Points	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how extrusion affects the integrity of fuel systems.
(b) Describe the steps to take if radioactivity is detected on an aircraft component.
(c) Why is strict adherence to explosive regulations critical for aircrew safety?

12. Home Assignment and Follow Up	Prepare a brief report on the importance of monitoring lubricant extrusion and exudation in aircraft maintenance, referencing relevant regulatory guidelines.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Categories of explosives, Various groups of explosives based on fire risk, explosion and hazard risk.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min